

G5: Local Geometry, Shared Structure, and Calibration

Recent discussion summary - seven encoders, two modalities, four text objectives

Core conclusion: Across all 21 encoder pairs, agreement is strongest locally and weakens toward the global CKA limit. The result appears in both nearest-neighbor overlap and CKNNA, while null calibration strengthens - rather than weakens - the observed objective-vs-lineage contrast.

Data used in G5

Spaces	img_small, img_base, img_large, txt_bge, txt_gpt2, txt_bert, txt_sbert
Common aligned rows	9,533
Neighborhood analysis	1,500 held-out items
Pair count	21 unordered encoder pairs
Primary local metric	k-nearest-neighbor overlap, reported raw and relative to chance
Scale sweep	CKNNA from small k to k = n-1, where it reaches the row-centered CKA limit

Scope note: this document summarizes the recent G1/G5 discussion and the supplied G5 outputs. It distinguishes measured results from interpretation and from proposed causal explanations.

1. What G5 is asking

G5 does not ask whether two models have the same coordinates. It asks whether they organize the same items similarly, and at what geometric scale that agreement is strongest.

Three distinct questions: Similarity = do two spaces preserve comparable relationships? Shared space = can those relationships be expressed in one coordinate frame? Geometry = what is the pattern of near/far relationships among items?

1.1 Global shape agreement

For a pair of encoders A and B, compare the rank ordering of pairwise distances among the same held-out items. A high Spearman rho means the two spaces preserve a similar global shape even before any map is fitted.

Measured group means from G1/G5 context: image-image rho = +0.561; text-text rho = +0.395; cross-modal rho = +0.291.

1.2 Local neighborhood agreement

For each item i and neighborhood size k , compute the overlap between the k nearest neighbors selected by encoder A and encoder B:

Definition $\text{overlap@k} = |N_k^A(i) \cap N_k^B(i)| / k$, averaged across items.

This asks a more local question: do the encoders agree on what is near what, even if their global layouts differ?

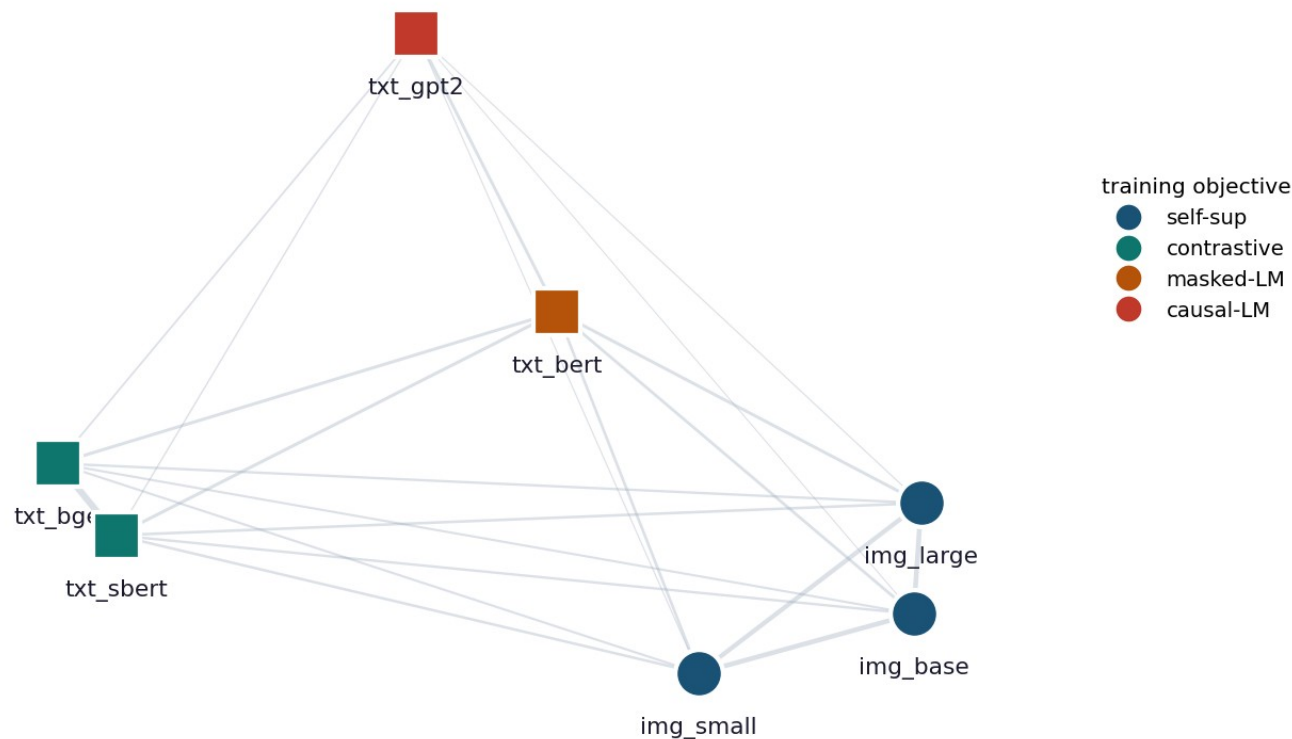
1.3 CKNNA as the local-to-global knob

CKNNA varies k continuously. Small k emphasizes local neighborhoods; as k grows, the metric becomes more global. In this run, $k = n-1$ is the row-centered CKA limit by construction.

Key G5 result All 21 of 21 pairs score higher at small k than at the global limit.

2. Encoder map: turning pairwise agreement into a picture

A map of the seven encoders
MDS on distance = $-\log(\text{shape agreement})$; edge width = agreement



Circles = image encoders, squares = text encoders. Proximity means the two spaces agree on SHAPE, not that their coordinates are similar — no map is fitted anywhere in this figure. Distance recipe follows Huh et al. 2024.

Figure 1. Seven encoders positioned by MDS using distance = $-\log(\text{shape agreement})$. Edge width encodes agreement.

How to read it: high shape agreement becomes a short MDS distance; low agreement becomes a long distance. The x/y axes themselves have no semantic meaning - only relative proximity matters.

- **bge-m3 and SBERT**: the strongest pair; same broad contrastive semantic objective, no shared model lineage.
- **DINOv2 small/base/large**: a visible family cluster, consistent with shared DINOv2 ancestry/distillation lineage.
- **Cross-modal links**: longer on average, but clearly non-random; image and text spaces already agree modestly on relative semantic structure without a fitted map.
- **GPT-2**: geometrically isolated from the strongest clusters, consistent with its low-effective-rank / highly anisotropic geometry.

Careful wording The map supports an observed ordering in this seven-model sample; it does not by itself prove a universal causal law that objective > lineage > modality.

3. Raw shape agreement: what exists before fitting

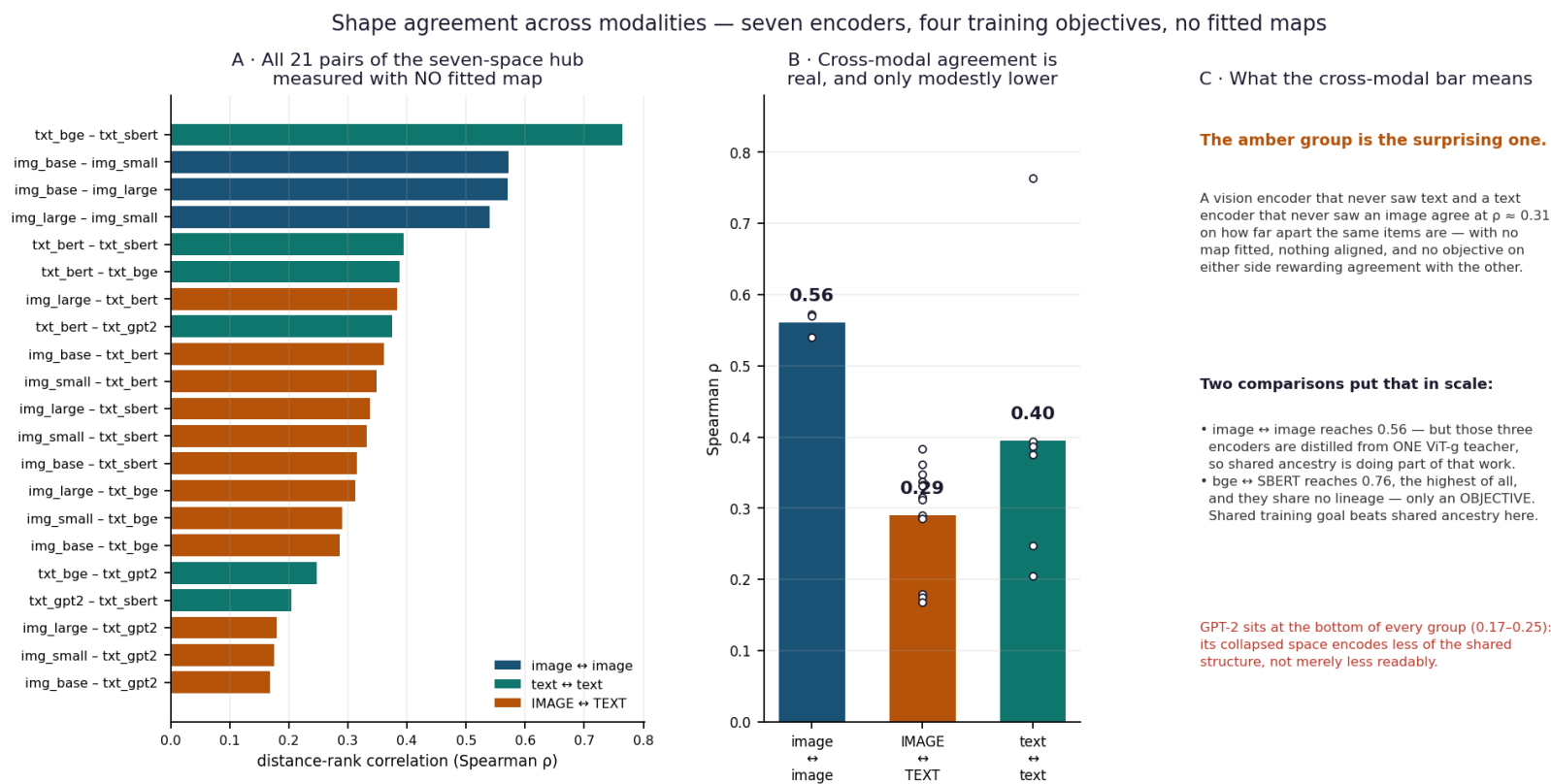
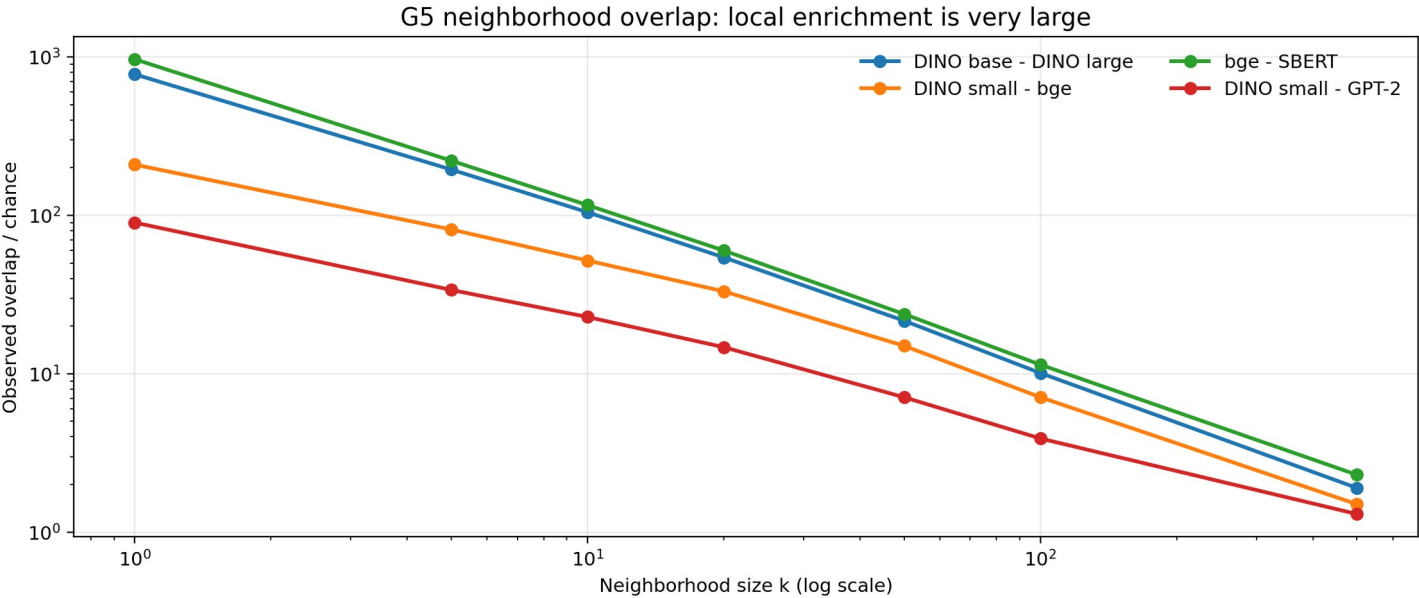


Figure 2. Pairwise distance-rank agreement across the seven spaces. No map is fitted for this test.

The cross-modal mean rho of about +0.29 is the surprising part: a vision encoder that never saw text and a text encoder that never saw an image still agree, to a meaningful degree, on how far apart the same semantic items are.

This is evidence that the Hub does not have to invent the correspondence from nothing. A correspondence already exists in raw geometry; the Hub mainly supplies a common coordinate system for using it.

4. Neighborhood overlap: the local signal



Caution: the ratio-to-chance has a k-dependent maximum (N/k), so its decline alone is not proof of local > global.

Figure 3. Representative G5 neighborhood-overlap enrichment relative to chance.

Examples at k=10:

DINO base - DINO large	0.697 raw overlap; 104.6x chance
bge-m3 - SBERT	0.773 raw overlap; 116.0x chance
DINO small - bge-m3	0.345 raw overlap; 51.8x chance

DINO small - GPT-2

0.152 raw overlap; 22.8x chance

Interpretation At small k , encoders agree far above chance about immediate semantic neighborhoods. Cross-modal agreement is weaker than within-family or same-objective agreement, but still substantial.

Important caveat The ratio-to-chance declines partly by construction because chance overlap rises with k and the maximum possible ratio is approximately N/k . Therefore the dramatic $967x \rightarrow 2.3x$ decline is descriptive, but it should not be used alone as proof that alignment is local. CKNNA provides the cleaner scale test.

5. CKNNA: the clean local-to-global test

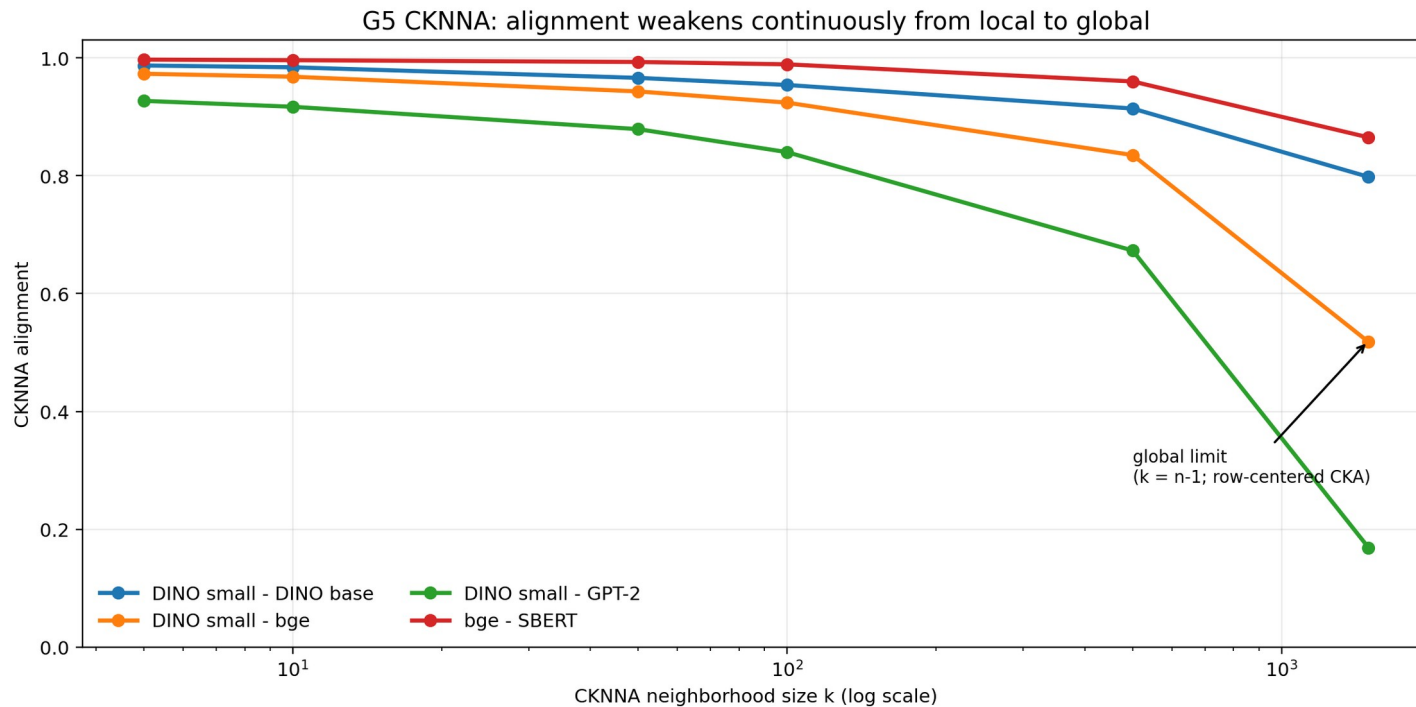


Figure 4. Representative CKNNA curves. Small k is local; $k = 1499$ is the global row-centered CKA limit.

Representative trajectories:

DINO small - DINO base0.987 at $k=5 \rightarrow 0.798$ globally

DINO small - bge-m3	0.973 -> 0.518
DINO small - GPT-2	0.927 -> 0.169
bge-m3 - SBERT	0.997 -> 0.865

The universal direction is the key result: 21/21 pairs are more aligned locally than at the global limit. This does not depend on one model, one modality pair, or one absolute score.

Why this is stronger than the overlap ratio CKNNA uses one metric family while varying the neighborhood scale. It therefore makes the local-to-global comparison without relying on a chance ratio whose upper bound changes with k.

6. What seems to drive local agreement?

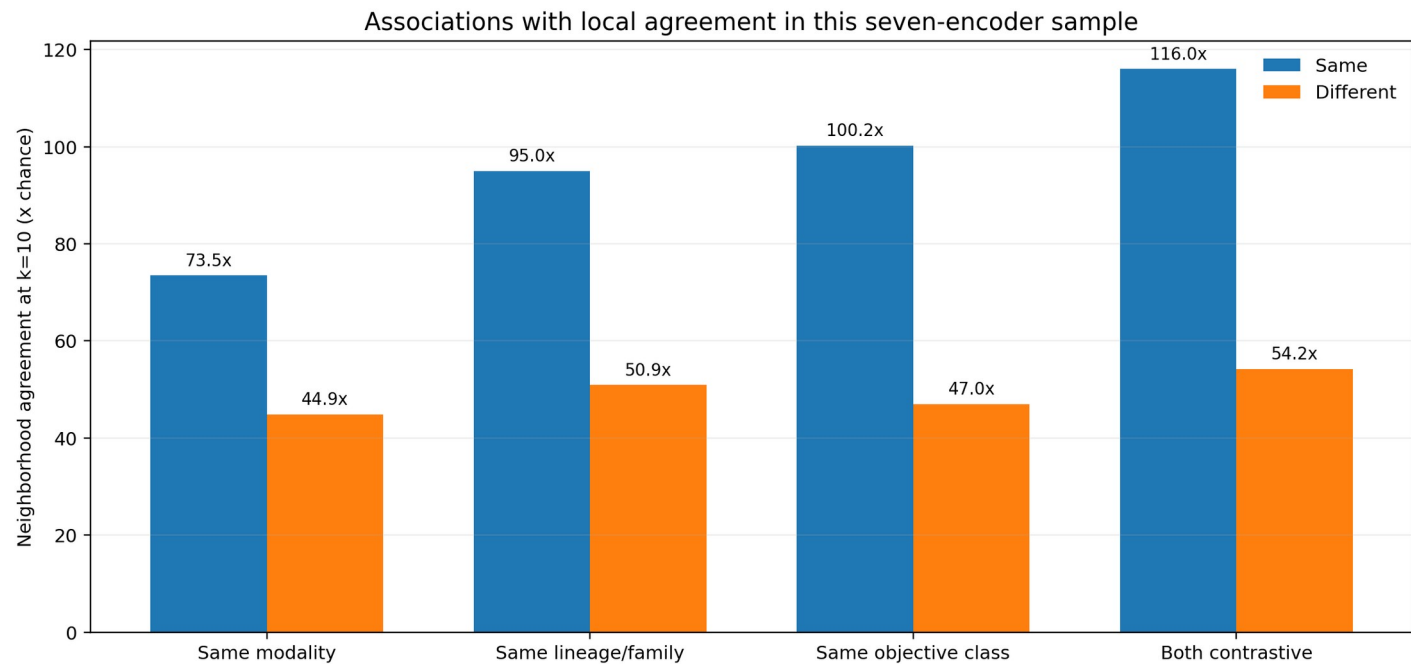


Figure 5. At k=10, shared objective shows the largest observed association with neighborhood agreement in this seven-encoder sample.

Same modality	73.5x chance vs 44.9x when modality differs; gap +28.6
Same lineage/family	95.0x vs 50.9x; gap +44.1
Same objective class	100.2x vs 47.0x; gap +53.2
Both contrastive	116.0x vs 54.2x; gap +61.8

The highest-agreeing pair is bge-m3 - SBERT at 116x chance for k=10. It shares a broad contrastive semantic objective but not lineage, which argues against the idea that all observed similarity is merely inherited ancestry.

Defensible conclusion Within these seven encoders, shared objective is associated with the strongest local neighborhood agreement, followed by lineage and modality. Because the number of independent families/objective classes is small, treat this as a strong exploratory ordering, not a universal causal hierarchy.

7. GPT-2: local information survives a poor global geometry

GPT-2 is consistently among the weakest raw-shape and neighborhood partners, yet the CKNNA sweep reveals an important nuance.

DINO small - GPT-2	CKNNA 0.927 at k=5 -> 0.169 globally
DINO base - GPT-2	0.917 -> 0.166
DINO large - GPT-2	0.903 -> 0.166
bge-m3 - GPT-2	0.950 -> 0.229

The local scores are high while the global scores collapse. This is consistent with the earlier geometry diagnosis: GPT-2 can retain useful local correspondence even when its global representation is highly anisotropic / low-rank and hard to use with cosine retrieval.

Read vs write asymmetry from G1 GPT-2 can be a strong SOURCE through the Hub but a poor TARGET. In the seven-space composition table, reading out of GPT-2 retained about 95.1% R@1 on average, while writing into GPT-2 retained only about 24.8%.

This supports a broader distinction: information may be present but geometrically difficult to read globally. Local-neighborhood measurements can expose structure that global metrics hide.

8. Null calibration and the width confound

The recent discussion considered a later calibration critique: raw global similarity metrics such as CKA can have a null baseline that changes with representation width. In the supplied follow-up, the CKA null increased with width (reported Spearman +0.56).

DINO base - DINO large (wide lineage pair)	raw CKA 0.887 -> calibrated 0.662; drop 0.225
bge-m3 - SBERT (same objective, narrower)	raw CKA 0.911 -> calibrated 0.843; drop 0.068
Raw gap	0.911 - 0.887 = +0.024
Calibrated gap	0.843 - 0.662 = +0.181 (about 7.5x wider)

The correction therefore does not explain away the objective-over-lineage contrast. It disproportionately removes inflation from the wider DINO lineage pair and makes the observed separation larger.

8.1 Why the primary kNN metric is safer

The supplied null check for neighborhood overlap was approximately +0.016, matching the expected k/N behavior. Unlike raw CKA, this null is controlled by neighborhood size and sample count rather than embedding width.

Best defense wording The primary neighborhood result is not a width artifact: its null follows the expected k/N form. Width calibration affects the global CKA comparison, and after correction the secondary objective-vs-lineage finding becomes stronger.

8.2 One correction to the rhetoric

Say "the width confound biased the comparison in favor of the wider DINO lineage pair" rather than "the confound was working against us." The former is precise about which comparison was affected.

9. Three metrics, one geometric story

Global distance-rank Spearman rho	Shared structure exists without fitting, but cross-modal global agreement is only
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moderate (mean about +0.29).

Nearest-neighbor overlap

Immediate neighborhoods overlap tens to hundreds of times chance at small k .

CKNNA(k)

For 21/21 pairs, alignment is higher locally than at the global CKA limit.

The methods are complementary: Spearman tests broad ordering of distances; k NN overlap asks whether the same neighbors are selected; CKNNA provides a continuous local-to-global similarity scale. Their agreement makes the conclusion much less metric-specific.

Refined scientific claim Representational convergence in this sample is strongest in local semantic neighborhoods and weaker in complete global geometry. The models need not be globally isomorphic to share a useful local semantic manifold.

9.1 Relationship to the shared Hub

G5 changes how to interpret G1. The 512-d Hub should not be described as creating a shared representation from nothing. The raw spaces already contain partial correspondence - especially locally. The Hub supplies a common linear coordinate frame that exposes, composes, and transports that pre-existing structure.

Conceptual chain raw geometry $\rightarrow$ local shared neighborhoods $\rightarrow$ common Hub coordinates $\rightarrow$ cross-encoder transport / reusable downstream head

10. Defense-ready statements

What did G5 add? It localized the convergence. G1 showed that seven spaces can be composed through one Hub; G5 shows that the correspondence is strongest in local neighborhoods rather than in one globally identical geometry.

Could the result be caused by embedding width? The primary k NN metric has the expected k/N null and does not inherit the width-dependent CKA baseline. When CKA is calibrated, the bge-SBERT objective pair separates even further from the wide DINO lineage pair.

Is the local result metric-specific? No. Nearest-neighbor overlap and CKNNA independently give the same ordering, and all 21 pairs have higher CKNNA locally than at the global limit.

Does objective beat lineage? In this seven-encoder sample, shared objective has the strongest association with local agreement, and the highest pair is bge-SBERT despite no shared lineage. This is strong exploratory evidence, not yet a universal causal law.

What does GPT-2 teach us? Local semantic correspondence can survive even when global geometry is collapsed. GPT-2 is therefore not simply information-poor; it is globally difficult to read and especially difficult to reconstruct into.

One-sentence conclusion

Across seven encoders, the same semantic neighborhoods recur much more reliably than the complete global geometry; this local structure survives width-aware calibration and is precisely the kind of pre-existing correspondence that a linear shared Hub can turn into usable common coordinates.

11. Compact result sheet

Dataset	7 spaces on 9,533 aligned common rows
G5 neighborhood sample	1,500 items
Raw shape agreement	image-image +0.561; text-text +0.395; image-text +0.291
Top raw-shape pair	bge-m3 - SBERT: rho +0.764
k=10 top neighborhood pair	bge-m3 - SBERT: 0.773 raw overlap; 116x chance
k=10 same objective class	100.2x chance vs 47.0x when objective differs
CKNNA local-to-global	21/21 pairs higher at small k than k=n-1
Representative cross-modal CKNNA	DINO small - bge: 0.973 (k=5) -> 0.518 global
Representative GPT-2 CKNNA	DINO small - GPT-2: 0.927 -> 0.169

G1 seven-space Hub	512-d balanced whitened PCA; one linear map per encoder
Hub composition	95.6% mean R@1 retention excluding structurally broken targets
GPT-2 Hub asymmetry	about 95.1% as source vs 24.8% as target
Head-transfer sweep	best at HUB_DIM=512, alpha=0.01; R@1 0.451; verification AUC 0.986

Prepared from the recent conversation, the supplied G5 run outputs, and the project status/defense materials. Numerical statements are intentionally kept tied to the reported run rather than generalized beyond the tested model set.